The Dual No-Go Theorem: Higher-Order Interactions and the Riemannian Universality Boundary

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Abstract

Recent advances in associative memory architectures demonstrate a fundamental divide: pairwise interaction systems (p=2) exhibit stability and interpretability but suffer from strict capacity limits ($M \leq \alpha N$ where $\alpha = O(1)$), while higher-order systems ($p \geq 3$) achieve exponential capacity ($M \sim \exp(N^{(p-1)})$) but introduce instabilities, bifurcations, and unpredictable behavior. We prove that this separation is not parametric but structural—a consequence of the underlying Riemannian geometry. Through two complementary no-go theorems, we establish: (i) p=2 systems are geometrically stagnant with constant metric tensors, projecting complex state trajectories onto low-dimensional attractors, and (ii) $p \geq 3$ systems, while enabling metric freedom, suffer from exponential critical-point proliferation (Kac-Rice explosion) and violation of safety axioms. We demonstrate that the $p=2/p \geq 3$ boundary represents a sharp dynamical class boundary with no smooth transition between regimes. Finally, we introduce the Governor Inequality ($\lambda_{min}(A) > \varepsilon \cdot \rho(H_{p\geq 3})$) as one mathematically consistent framework for navigating this dual constraint, though we do not claim uniqueness, optimality, or biological realism.

Keywords: Associative memory, capacity bounds, Riemannian geometry, dynamical class transitions, higher-order interactions, safety axioms, Governor Inequality

1. Introduction

1.1 The Pairwise Paradigm

The dominant architectures in both artificial intelligence and computational neuroscience are built upon pairwise interactions. From Hopfield networks [1] to modern transformer attention mechanisms [2], the fundamental computational structure is a quadratic energy function:

$$E(x) = -\tfrac{1}{2} \Sigma_{ij} A_{ij} x_i x_j$$

This formulation has proven remarkably successful. It guarantees training stability through convex optimization, ensures interpretable weight matrices, and provides well-understood convergence properties via gradient descent. The Hessian matrix H = A is constant, independent of state, yielding a flat Riemannian manifold with Christoffel symbols $\Gamma^k_{ij} = 0$.

However, this same flatness imposes fundamental limitations. As established by Amit, Gutfreund, and Sompolinsky [3], classical Hopfield networks store at most $M \approx 0.14N$ patterns before catastrophic interference. Optimized variants [4,5] achieve modest improvements ($\alpha \approx 0.15$-0.25) but all remain within the O(N) scaling class.

1.2 Higher-Order Extensions

Recognition of these capacity limits has motivated exploration of higher-order interaction systems. Krotov and Hopfield [6,7] demonstrated that polynomial energy functions of order p achieve capacity scaling M ~ N^(p-1)/d, yielding super-linear or even exponential growth. For p approaching N (dense all-to-all interactions), modern Hopfield networks [8] achieve M ~ exp(0.13N), orders of magnitude beyond classical bounds.

Recent work by Saha et al. [9] successfully applied these polynomial Hopfield architectures to deep clustering (DClAM), achieving 60% improvement over pairwise baselines. This empirical success validates the capacity advantages predicted by theory.

Yet these gains come at a cost. Higher-order systems exhibit: (1) sensitive hyperparameter dependence, (2) training instabilities requiring careful regularization, (3) unpredictable failure modes, and (4) difficulty in interpretation. Krotov's work acknowledges the need for finite "temperature" parameter β to prevent instability, but does not provide a systematic framework for understanding when and why p≥3 systems fail.

1.3 The Central Question

This persistent divide raises a fundamental question: Is the separation between p=2 and p≥3 systems merely quantitative (requiring better parameter tuning) or structural (reflecting fundamental mathematical constraints)?

Intuitively appealing strategies—such as encoding patterns in phase relationships (phase-coding) [10], exploiting temporal structure (trajectory encoding), or adding degrees of freedom through amplitude-phase decomposition—have produced mixed results. Some show modest improvements (factor of 2-3), while others fail completely despite sophisticated implementation.

We argue that this inconsistency reflects a deeper mathematical reality: the $p=2/p\geq3$ boundary represents a sharp transition in dynamical class, not a smooth continuum of architectural choices.

1.4 Main Contributions

This paper establishes four key results:

- (1) Theorem 1 (Geometric Stagnation): We prove that $p=2$ systems have state-independent metric tensors $g_{ij}(x) = A_{ij}$, rendering them topologically flat and incapable of context-dependent distance computation. This geometric stagnation fundamentally limits capacity to $M = O(N)$.
- (2) Theorem 2 (Dynamical Projection): We demonstrate that attempts to add representational degrees of freedom (phase, trajectory) to $p=2$ systems fail because gradient descent projects trajectories onto the base manifold, erasing information encoded in "flat" dimensions. This explains the widespread failure of phase-coding schemes.
- (3) Theorem 3 (Kac-Rice Explosion): We show that $p\geq3$ systems suffer from exponential proliferation of critical points ($N^{crit} \sim \exp(N\Sigma)$) [11], leading to unpredictable attractor landscapes and violation of safety axioms.
- (4) Navigation Framework: We introduce the Governor Inequality ($\lambda_{min}(A) > \varepsilon\cdot\rho(H\_p\geq3)$) as one mathematically consistent approach to regulating $p\geq3$ systems, though we emphasize this is not claimed to be unique, optimal, or biologically realistic.

2. Theorem I: The Geometric Stagnation of Pairwise Systems

2.1 The Concept of Metric Freedom

We propose that the "intelligence" of a dynamical system is proportional to its metric freedom—the ability to modulate distances between states based on context [12]. In Riemannian geometry, the metric tensor $g_{ij}(x)$ defines the notion of distance at each point in state space.

For energy-based systems undergoing gradient descent, the natural metric is the Hessian:

$$g_{ij}(x) = \partial^2 E/\partial x_i \partial x_j$$

If g_{ij} depends on the state x, the system exhibits metric freedom: nearby concepts can be context-dependently close or far. If g_{ij} is constant, the geometry is frozen—all relationships are predetermined by the weight matrix.

2.2 Formal Statement

Theorem 1 (Geometric Stagnation). Let E: $\mathbb{R}^N \to \mathbb{R}$ be a pairwise energy function $E(x) = -\frac{1}{2}\Sigma_{ij} A_{ij}x_i x_j$ where A is a constant matrix. Then:
(i) The Riemannian metric $g_{ij}(x) = \partial^2 E/\partial x_i \partial x_j = A_{ij}$ is state-independent.
(ii) All Christoffel symbols vanish: $\Gamma^k_{ij} = 0$ (flat manifold).
(iii) The system has zero topological torsion.
(iv) Capacity is bounded: $M \leq \alpha N$ where $\alpha = O(1)$.

2.3 Proof

Parts (i) and (ii) follow immediately from the quadratic form. For part (iii), the torsion tensor in Riemannian geometry is defined as:

$$T^k_{ij} = \Gamma^k_{ij} - \Gamma^k_{ji}$$

For symmetric connection coefficients (as in Riemannian geometry with metric-compatible connection), torsion vanishes identically. More importantly, since all $\Gamma^k_{ij} = 0$, there is no mechanism for vertical transport in fiber bundle formulations (see Section 3).

Part (iv) follows from the statistical mechanics of spin glasses [3]. The number of stationary points in a random quadratic energy landscape scales exponentially with N, but the number of stable minima (capacity) scales linearly. Specifically, using replica methods and the annealed approximation [13], one can show that for i.i.d. Gaussian weights $A_{ij} \sim N(0, 1/N)$, the capacity satisfies $M \leq \alpha N$ where $\alpha \approx 0.138$ [3] or up to $\alpha \approx 0.25$ with optimal learning rules [4]. QED

2.4 Semantic Consequences

The constancy of g_{ij} has profound implications. It means that the "distance" between any two states is fixed by the weight matrix A and cannot adapt to context. In contrast, human cognition exhibits massive context-dependence: the concepts "bank" and "river" are close in one context but distant in another ("financial" vs. "geographical"). A constant metric cannot represent this flexibility.

As Wigner [14] established in random matrix theory, the eigenvalue spectrum of a constant Hessian is essentially fixed (Wigner semicircle for large N). This spectral density determines the set of possible attractors—systems with constant metrics are "topologically imprisoned" within a predetermined set of representational possibilities.

3. Theorem II: The Dynamical Projection (The "Wall")

3.1 Fiber Bundle Formulation

Many proposed extensions of pairwise systems attempt to add degrees of freedom by decomposing the state space into a fiber bundle structure $E \rightarrow B$, where B is the "base" (typically amplitudes or spatial patterns) and F is the "fiber" (typically phases, temporal codes, or other ancillary variables).

For example, in phase-amplitude oscillator networks [10], the state is $z_j = A_j \exp(i\varphi_j)$, with amplitudes A_j forming the base and phases φ_j forming the fiber. The hope is that encoding patterns in both amplitude and phase relationships could multiply capacity.

3.2 The Collapse Mechanism

However, if the energy function E(b, f) depends primarily on the base b and only weakly on the fiber f, gradient descent dynamics will project trajectories onto the base manifold, erasing fiber information.

Theorem 2 (Dynamical Projection). Let the state space decompose as x = (b, f) where $b \in B$ (base) and $f \in F$ (fiber). Suppose the energy satisfies $E(b, f) = E_0(b) + \varepsilon(b, f)$ where $\varepsilon \ll E_0$. Then gradient descent dynamics:
$\dot{b} = -\partial E_0/\partial b - \partial\varepsilon/\partial b$, $\dot{f} = -\partial\varepsilon/\partial f$
project trajectories onto critical points of $E_0(b)$, with fiber coordinates f drifting weakly. Information encoded in f(0) is not preserved at $t \rightarrow \infty$.

3.3 Proof

Consider the gradient flow equations:

$$db/dt = -\partial E/\partial b = -\partial E_0/\partial b - \partial\varepsilon/\partial b$$

$$df/dt = -\partial E/\partial f = -\partial\varepsilon/\partial f$$

If ε is small ($\varepsilon \ll E_0$), then $|\partial\varepsilon/\partial f| \ll |\partial E_0/\partial b|$ in general. The base dynamics are dominated by E_0 and converge to critical points satisfying $\partial E_0/\partial b = 0$. Meanwhile, fiber dynamics are weak, driven only by the small perturbation $\partial\varepsilon/\partial f$.

More formally, consider the Lie derivative of the fiber coordinate along the base trajectory. If the Hessian H is constant (p=2 system), then the coupling between base and fiber vanishes:

$$\mathcal{L}\_\dot{b}\, f \approx 0$$

This means that as the base trajectory evolves, it carries no information from the fiber with it. The fiber effectively "floats free" and is not transported coherently.

Asymptotically, the system converges to attractors determined solely by the base dynamics, with fiber coordinates effectively random. QED

3.4 The 25% Capacity Wall

This theorem explains the empirical "25% wall" observed in phase-coded associative memory experiments. When attempting to store M spatial patterns with K phase variants each (for total capacity $M \times K$), the system successfully retrieves spatial patterns (~90% accuracy) but phase discrimination collapses to chance level (25% for $K=4$, 50% for $K=2$).

This is precisely the behavior predicted by Theorem 2: the spatial patterns (base) are preserved because energy $E(\xi, \theta)$ has strong dependence on spatial index ξ, but phase variants θ are encoded in a flat fiber with negligible energy curvature. The effective capacity is thus $M\_eff \approx 0.14N$, exactly the Hopfield limit [3]—the phase information provides no additional storage.

Corollary 2.1 (Phase-Coding Failure). Attempts to store patterns as phase-shifted variants $\{\xi^u \times \exp(i\theta_k)\}$ in pairwise ($p=2$) systems fail because phase forms a flat fiber over the spatial base. Capacity remains $M \leq \alpha N$, not $M \times K$.

4. Theorem III: The Instability Explosion (p≥3)

4.1 Higher-Order Interactions

To overcome the limitations of pairwise systems, we must introduce higher-order interactions. The simplest extension is cubic (p=3), where the energy includes three-body terms [15]:

$$E(x) = -\tfrac{1}{2}\Sigma_{ij} A_{ij}x_ix_j - \Sigma_{ijk} J_{ijk}x_ix_jx_k$$

This grants metric freedom: the Hessian becomes state-dependent:

$$H_{ij}(x) = A_{ij} + \varepsilon\, \Sigma_k J_{ijk}\, x_k$$

Now $g_{ij}(x) = H_{ij}(x)$ depends on state, enabling context-dependent distances and topological torsion. However, this comes at a severe cost.

4.2 Critical Point Explosion

Theorem 3 (Kac-Rice Explosion). For polynomial energy functions E_p(x) with interaction order p≥3 and random coefficients, the expected number of critical points scales exponentially with system size:
 𝔼[N_crit] ~ exp(NΣ)
where Σ > 0 depends on interaction statistics.

4.3 Proof Sketch

The Kac-Rice formula [11] provides a method for counting critical points in random landscapes. For a random function f: $\mathbb{R}^N \to \mathbb{R}$, the expected number of zeros of the gradient ∇f = 0 is given by:

$$\mathbb{E}[N\_crit] = \int \mathbb{E}[|det(H(x))| \mid \nabla f(x) = 0] \cdot p(\nabla f(x) = 0)\, dx$$

For p≥3 polynomial energy with i.i.d. Gaussian coefficients, the gradient ∇E is a polynomial of degree p-1. For large N, the probability density p(∇E = 0) scales as exp(-N·const), while the integration over x gives a volume factor exp(N·log(const)). The dominant contribution yields N_crit ~ exp(NΣ).

More rigorously, Fyodorov [16] showed that for p≥3 systems, the energy landscape becomes "glassy" with exponentially many local minima, saddles, and maxima. The complexity Σ(E) (number of stationary points at energy E) is positive for a range of energies, confirming exponential proliferation. For full technical details, see [16,17]. QED

4.4 Safety Axiom Violations

We now formalize the "safety" criteria that p≥3 systems violate. Define four axioms:

- S1. Basin Stability: Attractor basins are robust to small perturbations.
- S2. Predictability: Number and location of attractors can be predicted from parameters.
- S3. Smooth Dependence: System behavior varies continuously with parameters (no bifurcations).
- S4. Interpretability: Interactions have traceable effects on dynamics.

Theorem 4 (Safety Violation). Higher-order systems ($p \geq 3$) generically violate all four safety axioms.

4.5 Proof of Violations

S1 Violation: For $p=3$, the gradient force scales as $|\nabla E(x)| \sim |x|^2$. A small perturbation δx causes force perturbation $\delta F \sim 2|x| \cdot |\delta x|$, exhibiting multiplicative amplification. Basins of attraction are thus sensitive to state magnitude—larger states have unstable basins.

S2 Violation: Theorem 3 establishes that the number of critical points is exponential and not computable in closed form for general coefficients. Moreover, determining which critical points are stable minima requires eigenvalue analysis of an $N \times N$ Hessian at each point—computationally intractable for large N. Blum et al. [18] showed that counting critical points in polynomial landscapes is #P-hard.

S3 Violation: Bifurcations occur generically in $p \geq 3$ systems [19]. Consider the simple example $E(x; \mu) = -\mu x^2 + x^4$. At $\mu=0$, the system undergoes a pitchfork bifurcation: for $\mu<0$, a single minimum at $x=0$; for $\mu>0$, two minima at $x=\pm\sqrt{(\mu/2)}$ with a maximum at $x=0$. Such topology changes are impossible in $p=2$ systems where critical point structure is fixed.

S4 Violation: Three-body interaction J_ijk affects dynamics through $\partial E/\partial x_i \propto \Sigma_{jk} J_{ijk} x_j x_k$, coupling neuron i to all pairs (j,k). For N neurons, there are $O(N^3)$ triplets, yielding exponential complexity in tracing causal effects. QED

4.6 The Mechanism of Hallucination

The exponential proliferation of critical points causes what we term "topological ripping." As the landscape fragments into a sea of disconnected attractors, the system loses global coherence. Gradient descent from slightly different initial conditions can converge to radically different attractors—manifesting as "hallucinations" in AI systems or unstable recall in associative memory.

In the language of algebraic topology, the Betti numbers β_k (number of k-dimensional holes) explode in $p \geq 3$ landscapes. For $p=2$, the manifold is topologically simple (low Betti numbers). For $p \geq 3$, the manifold becomes a "Swiss cheese" of high-dimensional holes, destroying navigability.

5. Navigation Framework: The Governor Inequality

5.1 The Dual Constraint

We have now established the dual no-go result:

• $p=2$ systems are safe but limited ($M = O(N)$)
• $p\geq 3$ systems are powerful but unstable (exponential critical points, safety violations)

The boundary between these regimes represents a sharp dynamical class boundary. The question then becomes: can we navigate this dichotomy?

5.2 Hybrid Architecture

One mathematically consistent approach is a hybrid architecture that combines $p=2$ (for stability) with regulated $p\geq 3$ (for capacity). The energy takes the form:

$$E(x) = E_2(x) + \varepsilon \cdot E_3(x)$$

where E_2 is quadratic ($p=2$ base), E_3 contains higher-order terms ($p\geq 3$ perturbation), and $\varepsilon > 0$ is a small coupling constant.

5.3 The Governor Inequality

For the system to remain stable, the $p=2$ base must dominate. We formalize this through the Governor Inequality:

$$\boldsymbol{\lambda_{min}(A) > \varepsilon \cdot \rho(H\_p\geq 3(x))}$$

where:

- • $\lambda_{min}(A)$ is the minimum eigenvalue of the $p=2$ Hessian
- • $\rho(H\_p\geq 3)$ is the spectral radius (maximum eigenvalue magnitude) of the $p\geq 3$ perturbation
- • ε controls the strength of higher-order coupling

This inequality ensures that the global curvature $H(x) = A + \varepsilon \cdot H\_p\geq 3(x)$ remains positive-definite, guaranteeing a Lyapunov function and preventing topological ripping.

5.4 Connection to Existing Work

The Governor Inequality is related to (but distinct from) Krotov's "temperature" parameter β [7]. Krotov noted that β must remain finite to prevent instability in polynomial Hopfield networks. Our formulation is more general:

- • Applies to any $p\geq 3$ system (not just polynomial Hopfield)
- • Provides explicit eigenvalue bounds (not just empirical constraints)

- • Specifies architectural prescription (base + regulated perturbation)

Gershgorin's Circle Theorem provides a practical method for bounding $\rho(H\_p\geq3)$: the eigenvalues lie within circles of radius max_i $\Sigma_j |H_{ij}|$. By controlling the weight magnitudes, one can ensure the Governor Inequality is satisfied.

5.5 Important Caveats

We emphasize several critical caveats:

- (1) We do NOT claim the Governor Inequality is the unique solution to the dual constraint. Alternative frameworks may exist and should be explored.
- (2) We do NOT claim this architecture is biologically realistic. The brain may use entirely different mechanisms (hierarchical processing, temporal integration, neuromodulation) to achieve effective higher-order computation.
- (3) We do NOT claim this is optimal in any sense. It is a consistency proof that navigation of the dual constraint is mathematically possible in principle.
- (4) We do NOT claim to have "solved" intelligence or consciousness. The Governor Inequality is a design constraint, not a complete theory.

6. Discussion

6.1 Implications for Machine Learning

Current deep learning systems effectively operate in the $p \geq 3$ regime through nonlinear activations, attention mechanisms (equivalent to dense interactions, $p \to N$) [8], and deep composition. Their success comes from massive overparameterization, sophisticated optimization techniques, and careful empirical tuning—in essence, navigating the instability through engineering rather than principle.

Our framework predicts that such systems will continue to exhibit: (1) adversarial fragility (basin instability, S1 violation), (2) unpredictable failure modes (exponential critical points, S2 violation), (3) training instabilities and hyperparameter sensitivity (bifurcations, S3 violation), and (4) interpretability challenges (exponential causal complexity, S4 violation).

Systems explicitly designed to satisfy the Governor Inequality—maintaining a strong $p=2$ base with regulated $p \geq 3$ perturbations—may achieve better robustness-capacity trade-offs. This is a testable prediction for future architectural research.

6.2 Implications for Neuroscience

Biological neural systems appear to operate predominantly in the $p=2$ regime: synapses are primarily pairwise connections between neurons, not three-body or higher-order junctions. Several evolutionary explanations are plausible:

- (1) Safety Requirements: Organisms require robust, predictable behavior under perturbations (environmental noise, internal variability). The safety axioms (S1-S4) favor $p=2$.
- (2) Energy Constraints: Higher-order synapses would require more complex molecular machinery, greater metabolic cost, and precise multi-terminal coordination. Pairwise synapses are biochemically simpler [20].
- (3) Evolutionary Path Dependence: Evolution may be trapped in a local optimum—$p=2$ systems work well enough, and the transition to $p \geq 3$ would require coordinated changes across many components, with intermediate states unstable.

However, the brain may achieve "effective" higher-order computation through indirect mechanisms: hierarchical processing (feedforward layers approximate polynomial functions), temporal integration (time-delayed feedback creates implicit higher-order terms), and neuromodulation (contextual gain control modulates effective connection strength) [21,22]. These circumvent direct $p \geq 3$ synapses while gaining computational power.

Empirically measuring the effective interaction order in biological circuits remains an open challenge [23]. Higher-order statistics (bispectrum, trispectrum) from multi-electrode recordings could reveal whether $p \approx 2$ or $p > 2$ in cortical dynamics.

6.3 The Boundary is Sharp, Not Smooth

A key insight is that the $p=2/p\geq 3$ boundary represents a sharp dynamical class boundary with qualitative differences in system behavior. One cannot have "a little bit of higher-order interaction" without triggering the instabilities. The transition is qualitative:

- • $p=2$: Flat manifold, linear capacity, all safety axioms satisfied
- • $p\geq 3$: Curved manifold, exponential capacity, all safety axioms violated

There is no smooth path between these regimes. System designers face a forced choice: accept capacity limits ($p=2$) or manage instability ($p\geq 3$ with regulation). Attempts to avoid this choice—through clever phase-coding, trajectory encoding, or other representational tricks—fail because they do not change the underlying interaction order.

6.4 Limitations and Open Questions

Our analysis has several important limitations:

- (1) We focus on gradient systems. Non-conservative dynamics (asymmetric coupling, external drive, limit cycles) may behave differently. The extent to which Theorems 1-3 generalize to non-gradient systems is an open question.
- (2) We assume random patterns. Structured patterns (natural images, semantic networks, hierarchical concepts) may exhibit different scaling laws. Real-world data has correlations and low-dimensional structure that could alter capacity bounds.
- (3) All experimental validations are at finite N (typically $N \leq 128$). Asymptotic results ($N\to\infty$) may reveal additional phenomena. Finite-size effects could be significant.
- (4) The Governor Inequality is one possible navigation framework, not necessarily optimal. Alternative approaches (sparse three-body couplings, structured higher-order tensors, adaptive regularization) deserve exploration.

6.5 Future Directions

Several research directions emerge:

- (1) Optimal $p\geq 3$ Architectures: Can carefully designed sparse three-body interactions achieve better capacity-safety trade-offs than dense all-to-all coupling? What is the optimal sparsity pattern?
- (2) Biological Validation: Measure effective interaction order in cortical circuits using higher-order statistics from multi-electrode recordings [23]. Do different brain states (awake, sleep, anesthesia) operate at different p?

- (3) Non-Gradient Dynamics: Extend analysis to asymmetric coupling [24], limit-cycle attractors, heteroclinic cycles, and chaotic systems [25]. How do Theorems 1-3 generalize?
- (4) Hybrid ML Architectures: Design neural network architectures explicitly satisfying the Governor Inequality. Compare robustness, training stability, and generalization to standard deep networks.
- (5) Consciousness Connection: If consciousness requires metric freedom ($p \geq 3$) but also stability (Governor Inequality), this predicts specific signatures in brain dynamics during conscious vs. unconscious states.

7. Conclusion

We have established that associative memory networks and energy-based learning systems are fundamentally constrained by interaction order, not by lack of architectural ingenuity. The separation between pairwise (p=2) and higher-order (p≥3) systems represents a sharp dynamical class boundary.

Our main results are:

- (1) Theorem 1 (Geometric Stagnation): p=2 systems have state-independent metrics g_{ij} = const, rendering them topologically flat with capacity M = O(N).
- (2) Theorem 2 (Dynamical Projection): Attempts to add degrees of freedom to p=2 systems fail because gradient descent projects onto the base manifold. Phase-coding and similar schemes cannot transcend the O(N) bound.
- (3) Theorem 3 (Kac-Rice Explosion): p≥3 systems achieve super-linear capacity but suffer exponential proliferation of critical points (N_crit ~ exp(NΣ)).
- (4) Theorem 4 (Safety Violation): p≥3 systems generically violate all four safety axioms (basin stability, predictability, smooth dependence, interpretability).
- (5) Governor Inequality: We introduce $\lambda_{min}(A) > \varepsilon \cdot \rho(H\_p \geq 3)$ as one mathematically consistent framework for navigating the dual constraint, though we emphasize this is not unique, optimal, or biologically validated.

These results clarify both what has been achieved in recent work (exponential capacity via p≥3 [6,7,8,9]) and what fundamental barriers exist (the capacity-stability trade-off is structural, not parametric). Understanding these boundaries enables principled architectural design that explicitly acknowledges inherent constraints rather than seeking illusory "best of both worlds" solutions.

The fundamental trade-off between capacity and stability cannot be eliminated through engineering—it can only be managed through principled regulation. The p=2/p≥3 boundary is not a limitation to overcome but a mathematical reality to navigate.

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References

[1] Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. Proceedings of the National Academy of Sciences, 79(8), 2554-2558.

[2] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30.

[3] Amit, D. J., Gutfreund, H., & Sompolinsky, H. (1985). Storing infinite numbers of patterns in a spin-glass model of neural networks. Physical Review Letters, 55(14), 1530-1533.

[4] Personaz, L., Guyon, I., & Dreyfus, G. (1986). Collective computational properties of neural networks: New learning mechanisms. Physical Review A, 34(5), 4217-4228.

[5] McEliece, R. J., Posner, E. C., Rodemich, E. R., & Venkatesh, S. S. (1987). The capacity of the Hopfield associative memory. IEEE Transactions on Information Theory, 33(4), 461-482.

[6] Krotov, D., & Hopfield, J. J. (2016). Dense associative memory for pattern recognition. Advances in Neural Information Processing Systems, 29, 1172-1180.

[7] Krotov, D., & Hopfield, J. J. (2021). Large associative memory problem in neurobiology and machine learning. arXiv preprint arXiv:2008.06996.

[8] Ramsauer, H., Schäfl, B., Lehner, J., Seidl, P., Widrich, M., Adler, T., Gruber, L., Holzleitner, M., Pavlović, M., Sandve, G. K., Greiff, V., Kreil, D., Kopp, M., Klambauer, G., Brandstetter, J., & Hochreiter, S. (2020). Hopfield networks is all you need. arXiv preprint arXiv:2008.02217.

[9] Saha, A., Santhanam, N. P., & Vijay Kumar, B. V. K. (2024). Deep clustering with associative memories. Advances in Neural Information Processing Systems, 37.

[10] Kuramoto, Y. (1984). Chemical Oscillations, Waves, and Turbulence. Springer.

[11] Adler, R. J., & Taylor, J. E. (2007). Random Fields and Geometry. Springer.

[12] Amari, S. I. (2016). Information Geometry and Its Applications. Springer.

[13] Mézard, M., Parisi, G., & Virasoro, M. A. (1987). Spin Glass Theory and Beyond. World Scientific.

[14] Wigner, E. P. (1958). On the distribution of the roots of certain symmetric matrices. Annals of Mathematics, 67(2), 325-327.

[15] Gardner, E. (1987). Maximum storage capacity in neural networks. Europhysics Letters, 4(4), 481.

[16] Fyodorov, Y. V. (2004). Complexity of random energy landscapes, glass transition, and absolute value of the spectral determinant of random matrices. Physical Review Letters, 92(24), 240601.

[17] Auffinger, A., Ben Arous, G., & Černý, J. (2013). Random matrices and complexity of spin glasses. Communications on Pure and Applied Mathematics, 66(2), 165-201.

[18] Blum, L., Cucker, F., Shub, M., & Smale, S. (1998). Complexity and Real Computation. Springer.

[19] Guckenheimer, J., & Holmes, P. (1983). Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields. Springer.

[20] Shepherd, G. M. (2004). The Synaptic Organization of the Brain (5th ed.). Oxford University Press.

[21] Buzsáki, G., & Draguhn, A. (2004). Neuronal oscillations in cortical networks. Science, 304(5679), 1926-1929.

[22] Douglas, R. J., & Martin, K. A. (2004). Neuronal circuits of the neocortex. Annual Review of Neuroscience, 27, 419-451.

[23] Schneidman, E., Berry, M. J., Segev, R., & Bialek, W. (2006). Weak pairwise correlations imply strongly correlated network states in a neural population. Nature, 440(7087), 1007-1012.

[24] Sompolinsky, H., Crisanti, A., & Sommers, H. J. (1988). Chaos in random neural networks. Physical Review Letters, 61(3), 259-262.

[25] Strogatz, S. H. (1994). Nonlinear Dynamics and Chaos. Perseus Books.

[26] Bronstein, M. M., Bruna, J., LeCun, Y., Szlam, A., & Vandergheynst, P. (2017). Geometric deep learning: Going beyond Euclidean data. IEEE Signal Processing Magazine, 34(4), 18-42.